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TCA785HKLA1

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SIEMENS

Phase Control IC

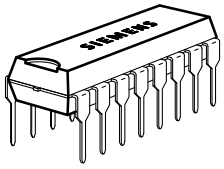
TCA 785

Pb-free lead plating; RoHS compliant

Bipolar IC

Features

-
- Large application scope
- May be used as zero point switch
-
- Three-phase operation possible (3 ICs)
- Output current 250 mA
- Large ramp current range
- Wide temperature range

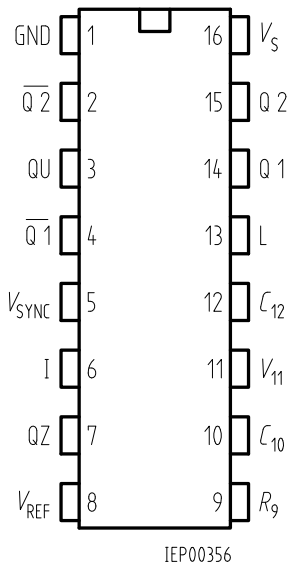


PG-DIP-16-1

Type	Ordering Code	Package
TCA 785	Q67000-A2321	PG-DIP-16-1

This phase control IC is intended to control thyristors, triacs, and transistors. The trigger pulses can be shifted within a phase angle between 0 ˚ and 180 ˚. Typical applications include converter circuits, AC controllers and three-phase current controllers.

This IC replaces the previous types TCA 780 and TCA 780 D.



Pin Configuration
(top view)

Pin Definitions and Functions

Pin	Symbol	Function
1	GND	Ground
2 3 4	$\overline{Q2}$ Q U $\overline{Q2}$	Output 2 inverted Output U Output 1 inverted
5	V_{SYNC}	Synchronous voltage
6 7	I Q Z	Inhibit Output Z
8	V_{REF}	Stabilized voltage
9 10	R_9 C_{10}	Ramp resistance Ramp capacitance
11	V_{11}	Control voltage
12	C_{12}	Pulse extension
13	L	Long pulse
14 15	Q 1 Q 2	Output 1 Output 2
16	V_S	Supply voltage

Functional Description

The synchronization signal is obtained via a high-ohmic resistance from the line voltage (voltage V_5). A zero voltage detector evaluates the zero passages and transfers them to the synchronization register.

This synchronization register controls a ramp generator, the capacitor C_{10} of which is charged by a constant current (determined by R_9). If the ramp voltage V_{10} exceeds the control voltage V_{11} (triggering angle φ), a signal is processed to the logic. Dependent on the magnitude of the control voltage V_{11}, the triggering angle φ can be shifted within a phase angle of 0° to 180°.

For every half wave, a positive pulse of approx. 30 µs duration appears at the outputs Q 1 and Q 2. The pulse duration can be prolonged up to 180° via a capacitor C_{12}. If pin 12 is connected to ground, pulses with a duration between φ and 180° will result.

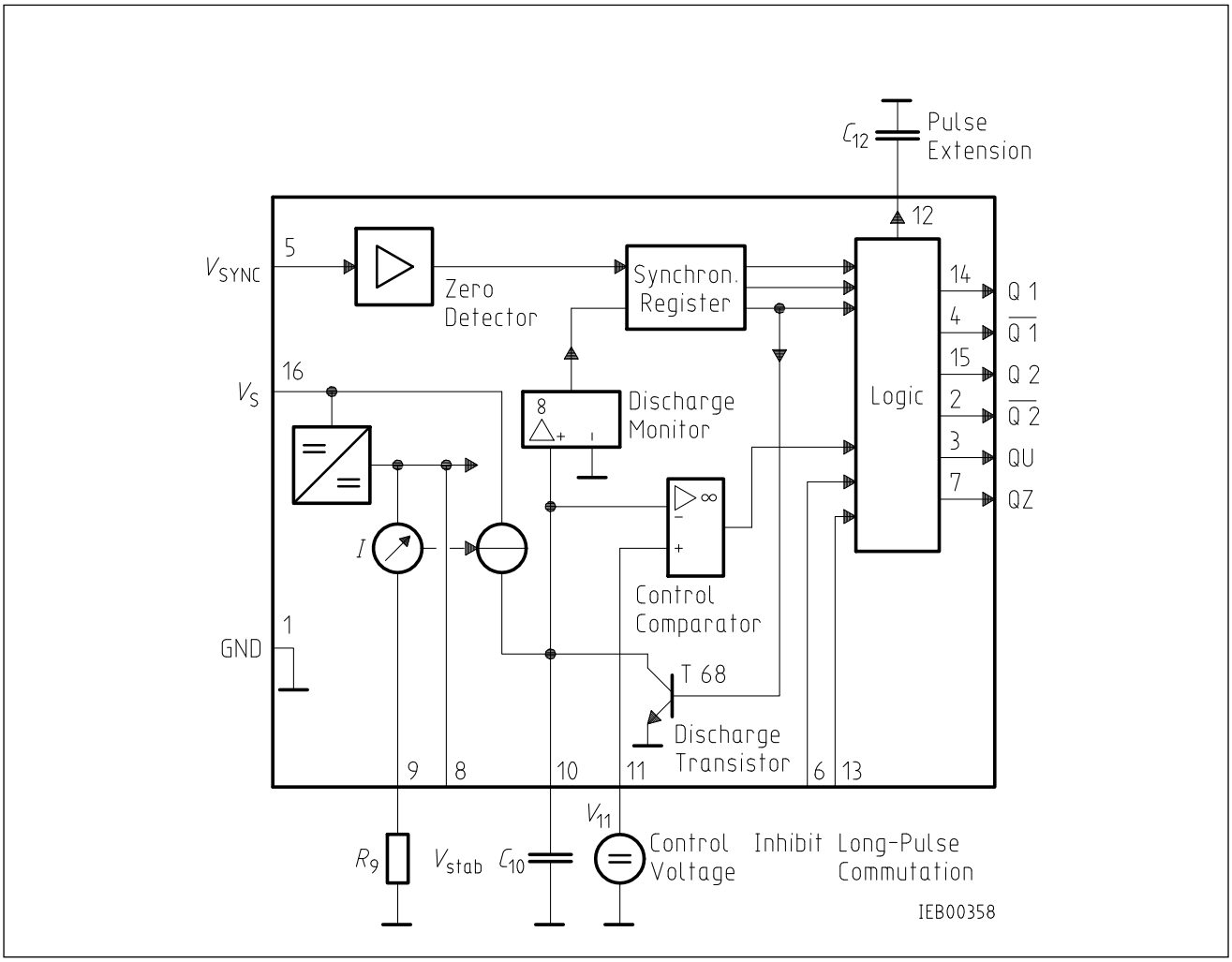
Outputs $\overline{Q1}$ and $\overline{Q2}$ supply the inverse signals of Q 1 and Q 2.

A signal of φ +180° which can be used for controlling an external logic,is available at pin 3.

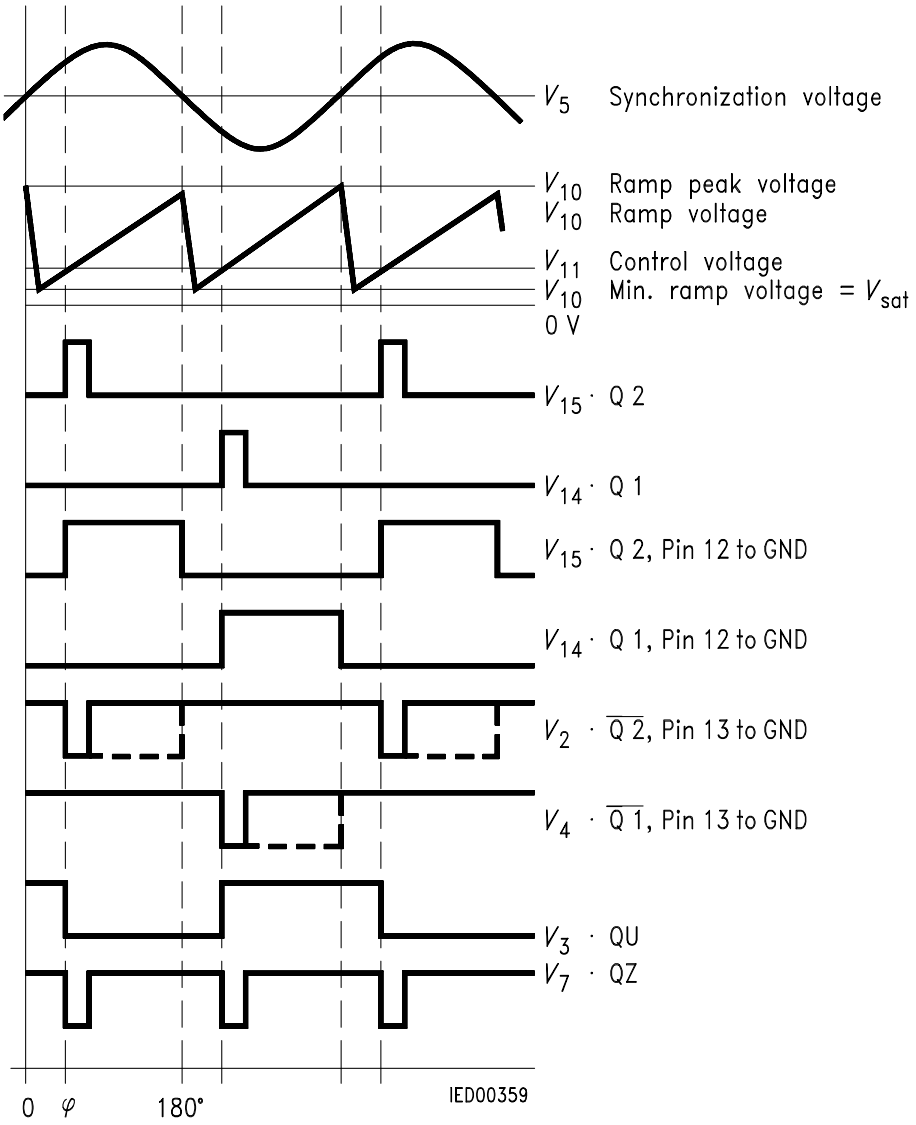
A signal which corresponds to the NOR link of Q 1 and Q 2 is available at output Q Z (pin 7).

The inhibit input can be used to disable outputs Q1, Q2 and $\overline{Q1}$, $\overline{Q2}$.

Pin 13 can be used to extend the outputs $\overline{Q1}$ and $\overline{Q2}$ to full pulse length (180° – φ).



Block Diagram



Pulse Diagram

Absolute Maximum Ratings

Parameter	Symbol	Limit Values		Unit
		min.	max.	
Supply voltage	V_S	– 0.5	18	V
Output current at pin 14, 15	I_Q	– 10	400	mA
Inhibit voltage Control voltage Voltage short-pulse circuit	V_6 V_{11} V_{13}	– 0.5 – 0.5 – 0.5	V_S V_S V_S	V V V
Synchronization input current	V_5	– 200	± 200	µA
Output voltage at pin 14, 15	V_Q		V_S	V
Output current at pin 2, 3, 4, 7	I_Q		10	mA
Output voltage at pin 2, 3, 4, 7	V_Q		V_S	V
Junction temperature Storage temperature	T_j T_{stg}	 – 55	150 125	˚C ˚C
Thermal resistance system - air	$R_{th\ SA}$		80	K/W

Operating Range

Supply voltage	V_S	8	18	V
Operating frequency	f	10	500	Hz
Ambient temperature	T_A	– 25	85	˚C

Characteristics

8 ≤ V_S ≤ 18 V; – 25 ˚C ≤ T_A ≤ 85 ˚C; f = 50 Hz

Parameter	Symbol	Limit Values			Unit	Test Circuit
		min.	typ.	max.		
Supply current consumption S1 … S6 open V_{11} = 0 V C_{10} = 47 nF; R_9 = 100 kΩ	I_S	4.5	6.5	10	mA	1
Synchronization pin 5 Input current R_2 varied Offset voltage	 $I_{5\ rms}$ ΔV_5	 30 	 30	 200 75	 µA mV	 1 4
Control input pin 11 Control voltage range Input resistance	 V_{11} R_{11}	 0.2 	 15	 $V_{10\ peak}$ 	 V kΩ	 1 5

Characteristics (cont'd)
$8 \le V_S \le 18$ V; $-25\ ^\circ\text{C} \le T_A \le 85\ ^\circ\text{C}$; $f = 50$ Hz

Parameter	Symbol	Limit Values			Unit	Test Circuit
		min.	typ.	max.		
Ramp generator						
Charge current	I_{10}	10		1000	µA	
Max. ramp voltage	V_{10}			$V_2 - 2$	V	1
Saturation voltage at capacitor	V_{10}	100	225	350	mV	1.6
Ramp resistance	R_9	3		30	kΩ	1
Sawtooth return time	t_f		80		µs	1
Inhibit pin 6 switch-over of pin 7						
Outputs disabled	$V_{6\,L}$		3.3	2.5	V	1
Outputs enabled	$V_{6\,H}$	4	3.3		V	1
Signal transition time	t_r	1		5	µs	1
Input current $V_6 = 8$ V	$I_{6\,H}$		500	800	µA	1
Input current $V_6 = 1.7$ V	$-I_{6\,L}$	80	150	20	µA	1
Deviation of I_{10} R_9 = const. $V_S = 12$ V; $C_{10} = 47$ nF	I_{10}	– 5		5	%	1
Deviation of I_{10} R_9 = const. $V_S = 8$ V to 18 V	I_{10}	– 20		20	%	1
Deviation of the ramp voltage between 2 following half-waves, V_S = const.	$\Delta V_{10\,max}$		± 1		%	
Long pulse switch-over pin 13 switch-over of S8						
Short pulse at output	$V_{13\,H}$	3.5	2.5		V	1
Long pulse at output	$V_{13\,L}$		2.5	2	V	1
Input current $V_{13} = 8$ V	$I_{13\,H}$			10	µA	1
Input current $V_{13} = 1.7$ V	$-I_{13\,L}$	45	65	10	µA	1
Outputs pin 2, 3, 4, 7 Reverse current $V_Q = V_S$	I_{CEO}			10	µA	2.6
Saturation voltage $I_Q = 2$ mA	V_{sat}	0.1	0.4	2	V	2.6

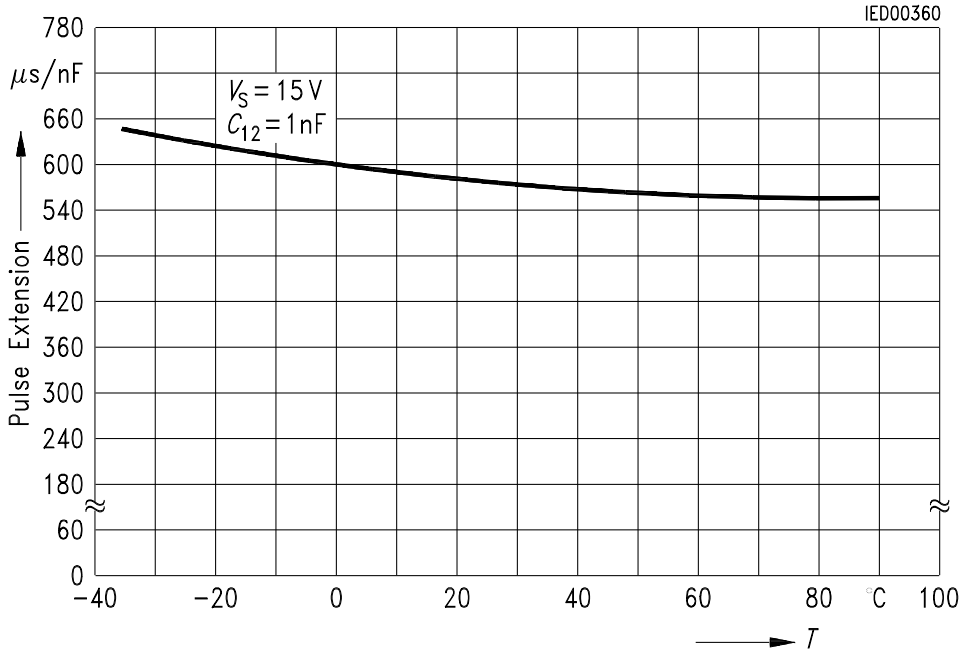
Characteristics (cont'd)
$8 \le V_S \le 18$ V; $-25\ ^\circ\text{C} \le T_A \le 85\ ^\circ\text{C}$; $f = 50$ Hz

Parameter	Symbol	Limit Values			Unit	Test Circuit
		min.	typ.	max.		
Outputs pin 14, 15						
H-output voltage $-I_Q = 250$ mA	$V_{14/15\,H}$	$V_S - 3$	$V_S - 2.5$	$V_S - 1.0$	V	3.6
L-output voltage $I_Q = 2$ mA	$V_{14/15\,L}$	0.3	0.8	2	V	2.6
Pulse width (short pulse) S9 open	t_p	20	30	40	µs	1
Pulse width (short pulse) with C_{12}	t_p	530	620	760	µs/nF	1
Internal voltage control						
Reference voltage Parallel connection of 10 ICs possible	V_{REF}	2.8 1	3.1	3.4	V	
TC of reference voltage	α_{REF}		2×10^{-4}	5×10^{-4}	1/K	1

Application Hints for External Components

		min	max	
Ramp capacitance	C_{10}	500 pF	1 µF[1]	The minimum and maximum values of I_{10} are to be observed
Triggering point	$t_{Tr} = \frac{V_{11} \times R_9 \times C_{10}}{V_{REF} \times K}$		[2]	
Charge current	$I_{10} = \frac{V_{REF} \times K}{R_9}$		[2]	Ramp voltage $V_{10\,max} = V_S - 2\ V$ $V_{10} = \frac{V_{REF} \times K \times t}{R_9 \times C_{10}}$ [2]

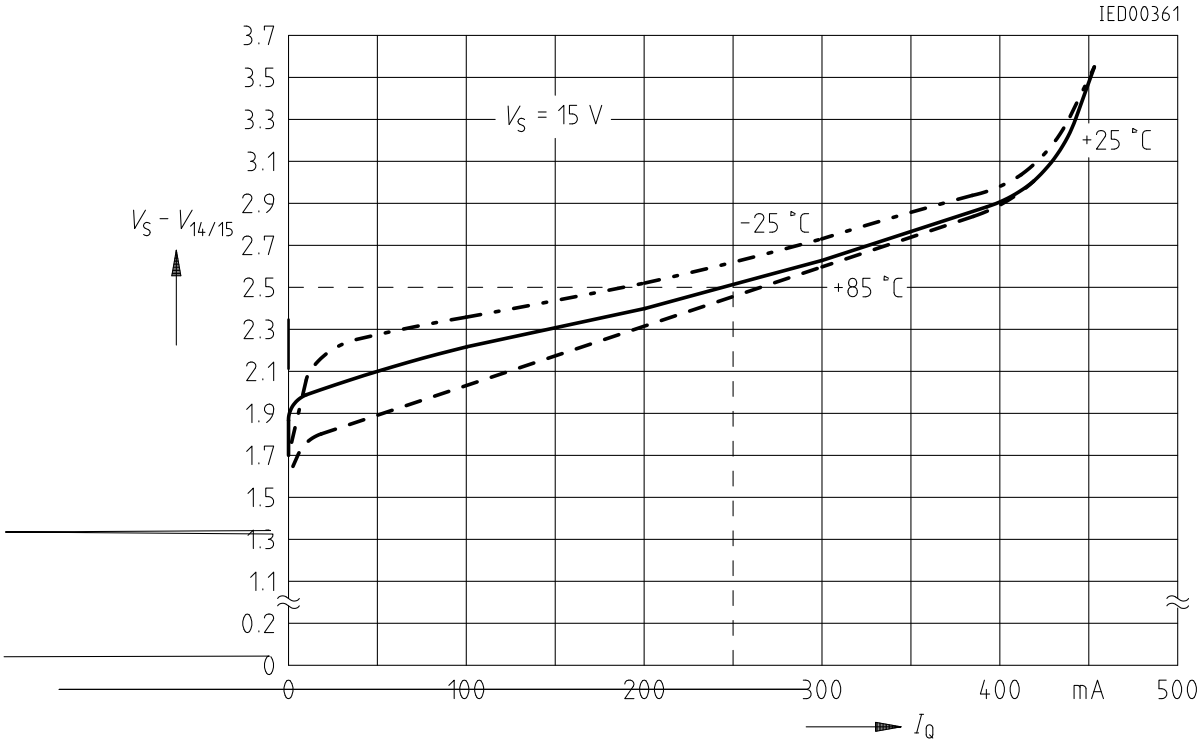
Pulse Extension versus Temperature



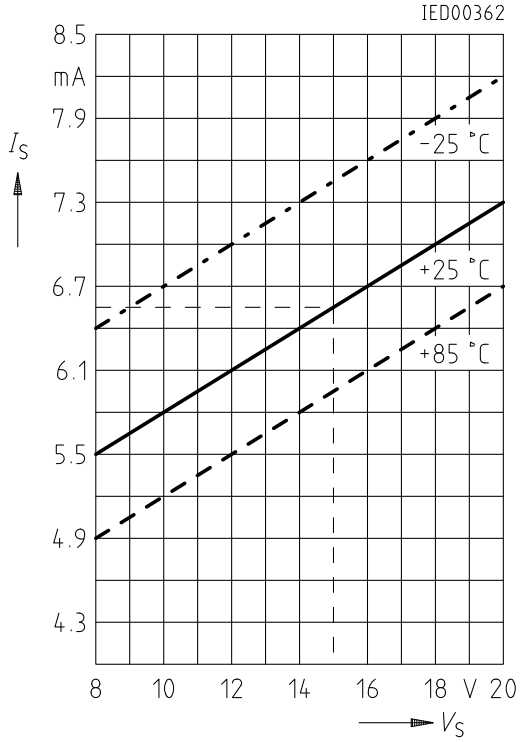
[1] Attention to flyback times

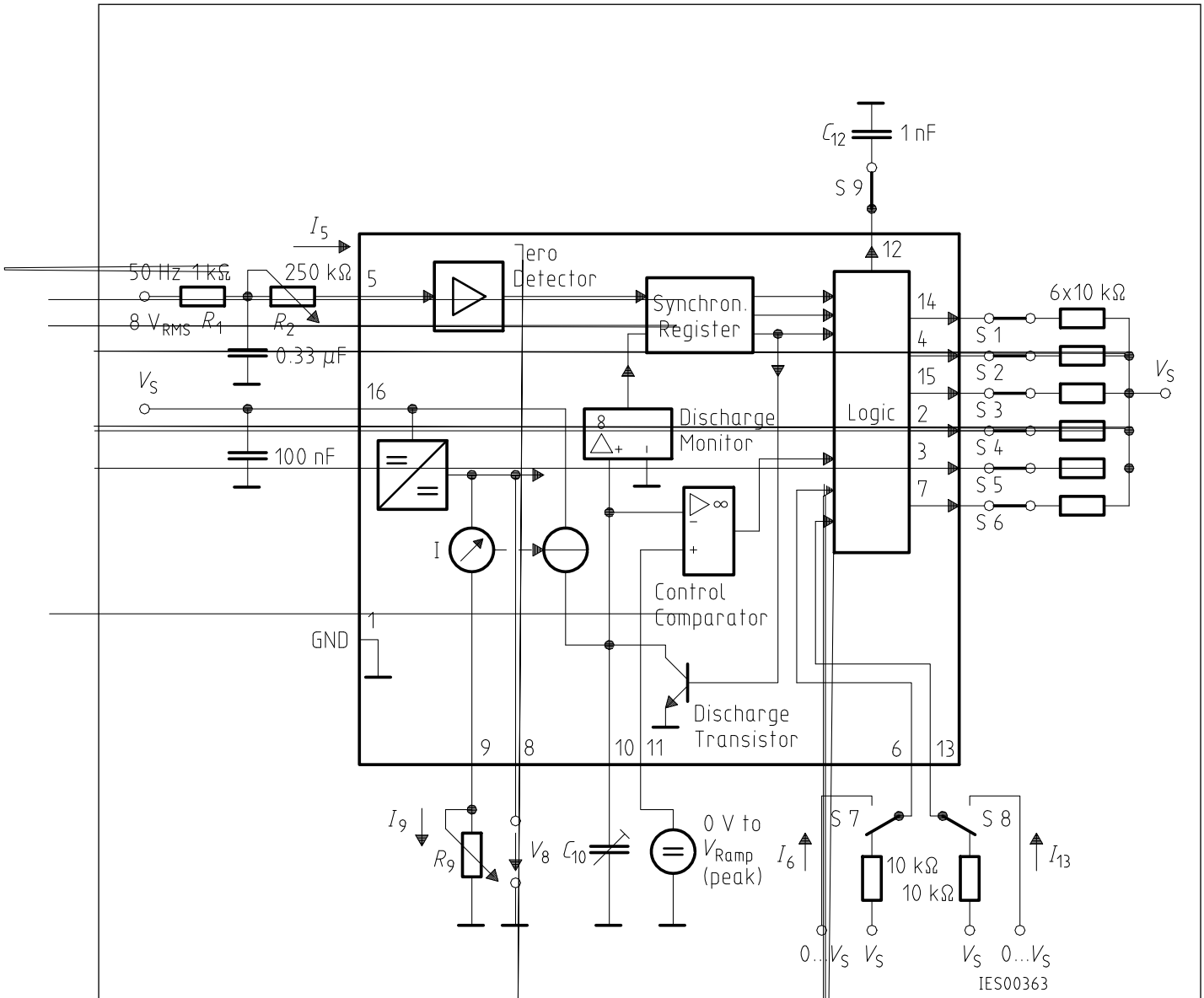
[2] K = 1.10 ± 20 %

Output Voltage measured to + V_S



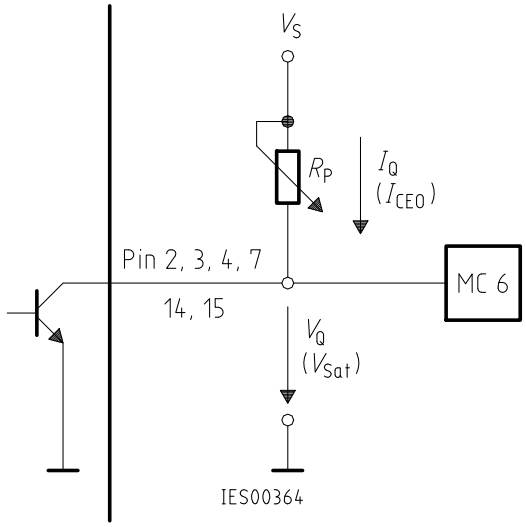
Supply Current versus Supply Voltage





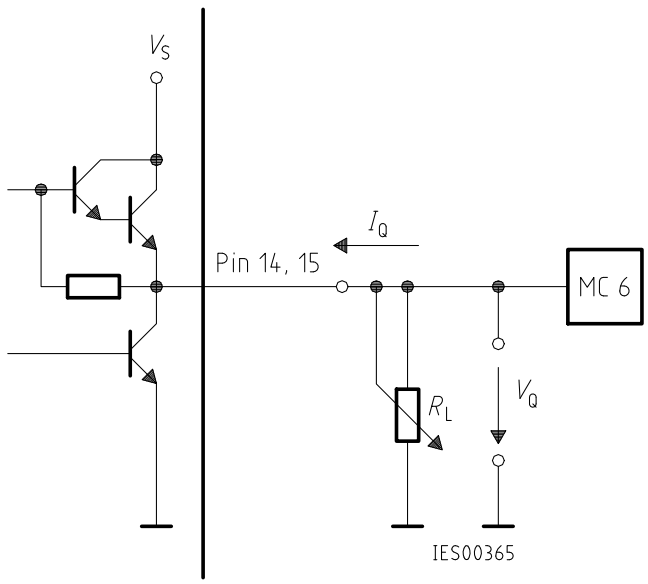
It is necessary for all measurements to adjust the ramp with the aid of C_{10} and R_9 in the way that $3\,V \leq V_{ramp\ max} \leq V_S - 2\,V$
e.g. C_{10} = 47 nF; 18 V: R_9 = 47 kΩ; 8 V: R_9 = 120 kΩ

Test Circuit 1



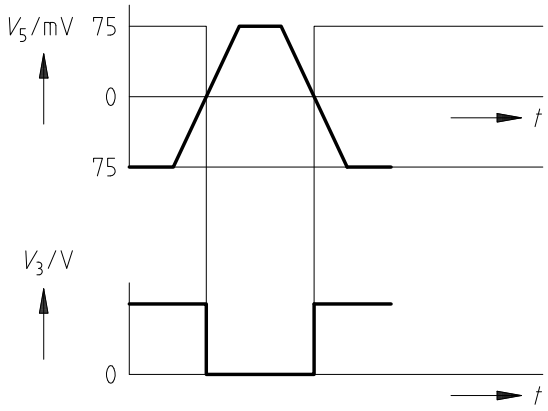
The remaining pins are connected as in test circuit 1

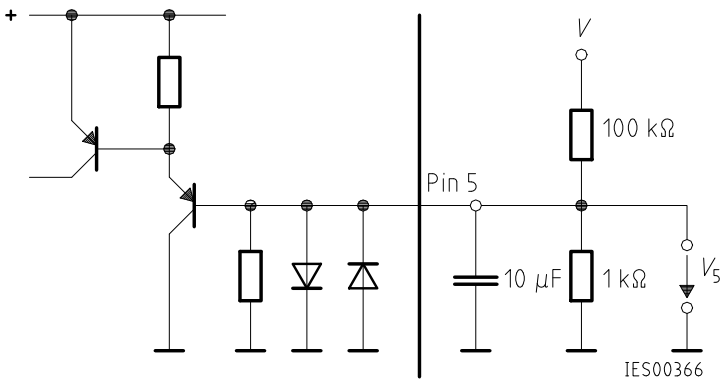
Test Circuit 2



The remaining pins are connected as in test circuit 1

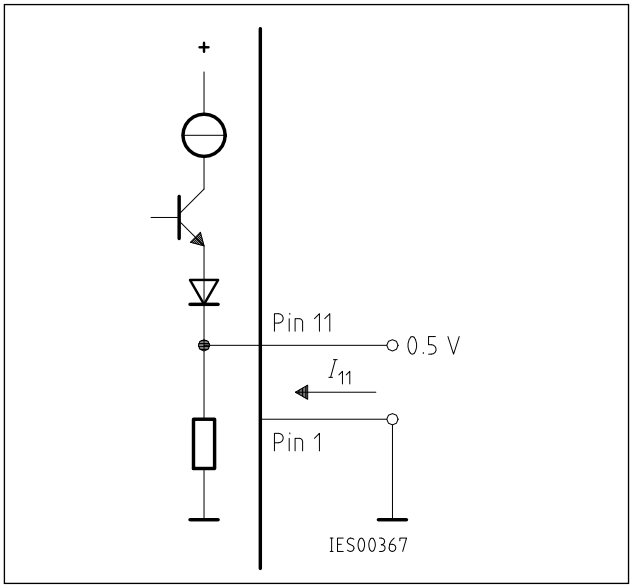
Test Circuit 3



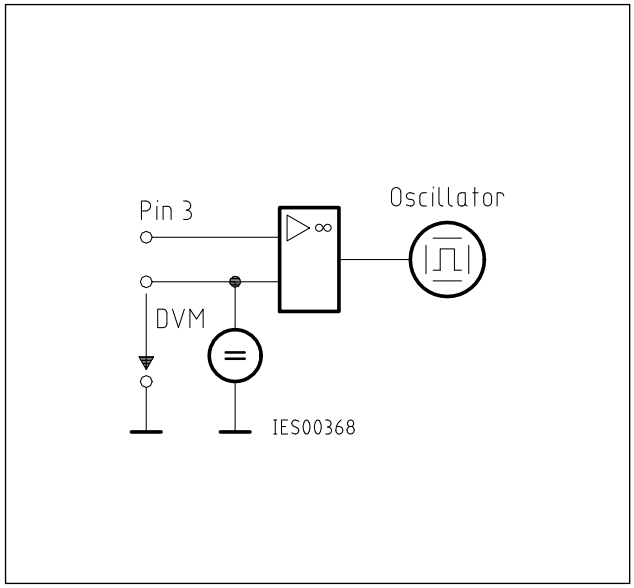


Remaining pins are connected as in test circuit 1
The 10 µF capacitor at pin 5 serves only for test purposes

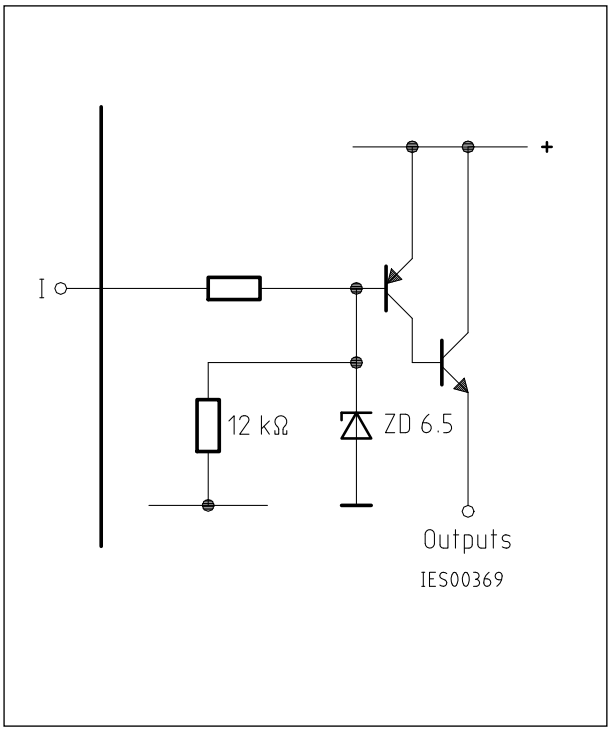
Test Circuit 4



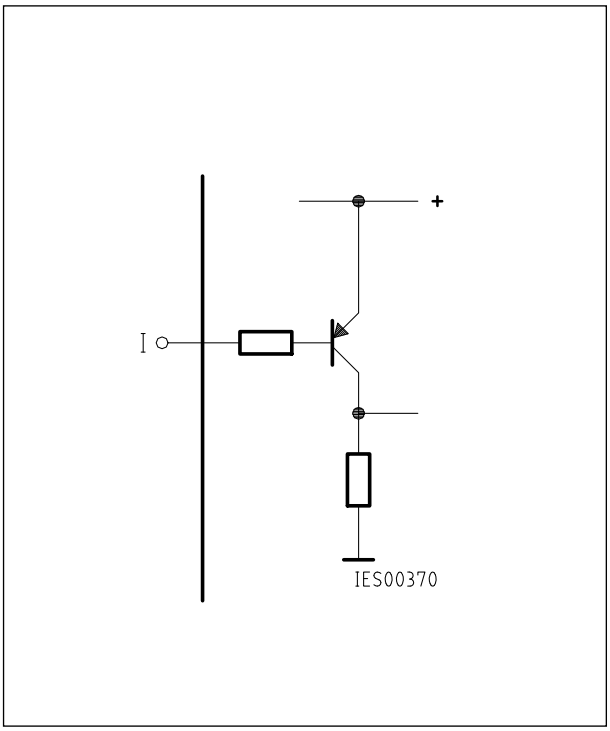
Test Circuit 5



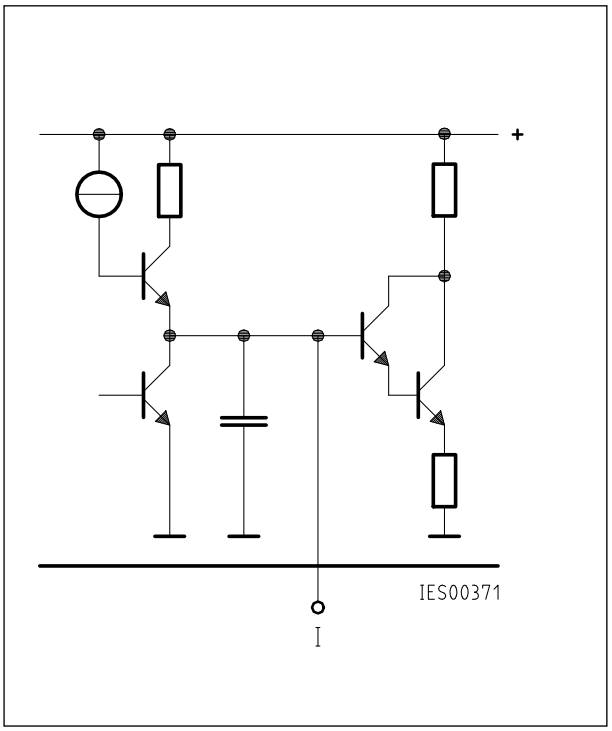
Test Circuit 6



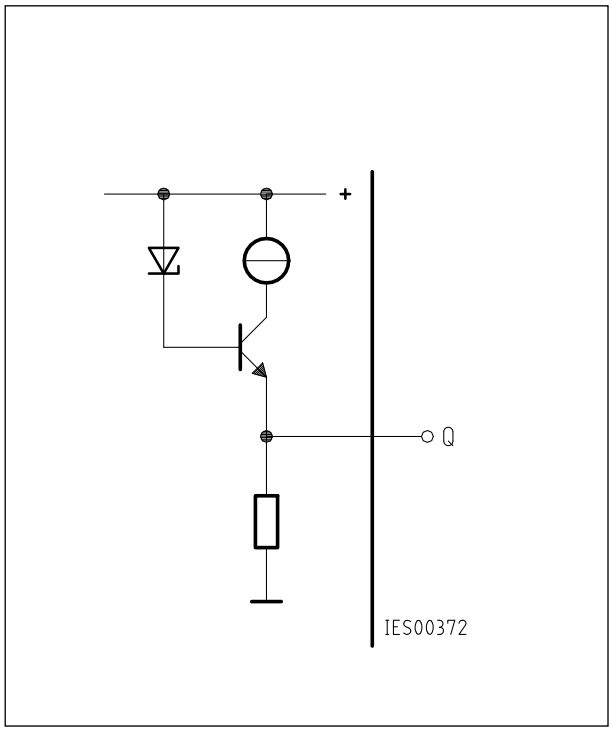
Inhibit 6



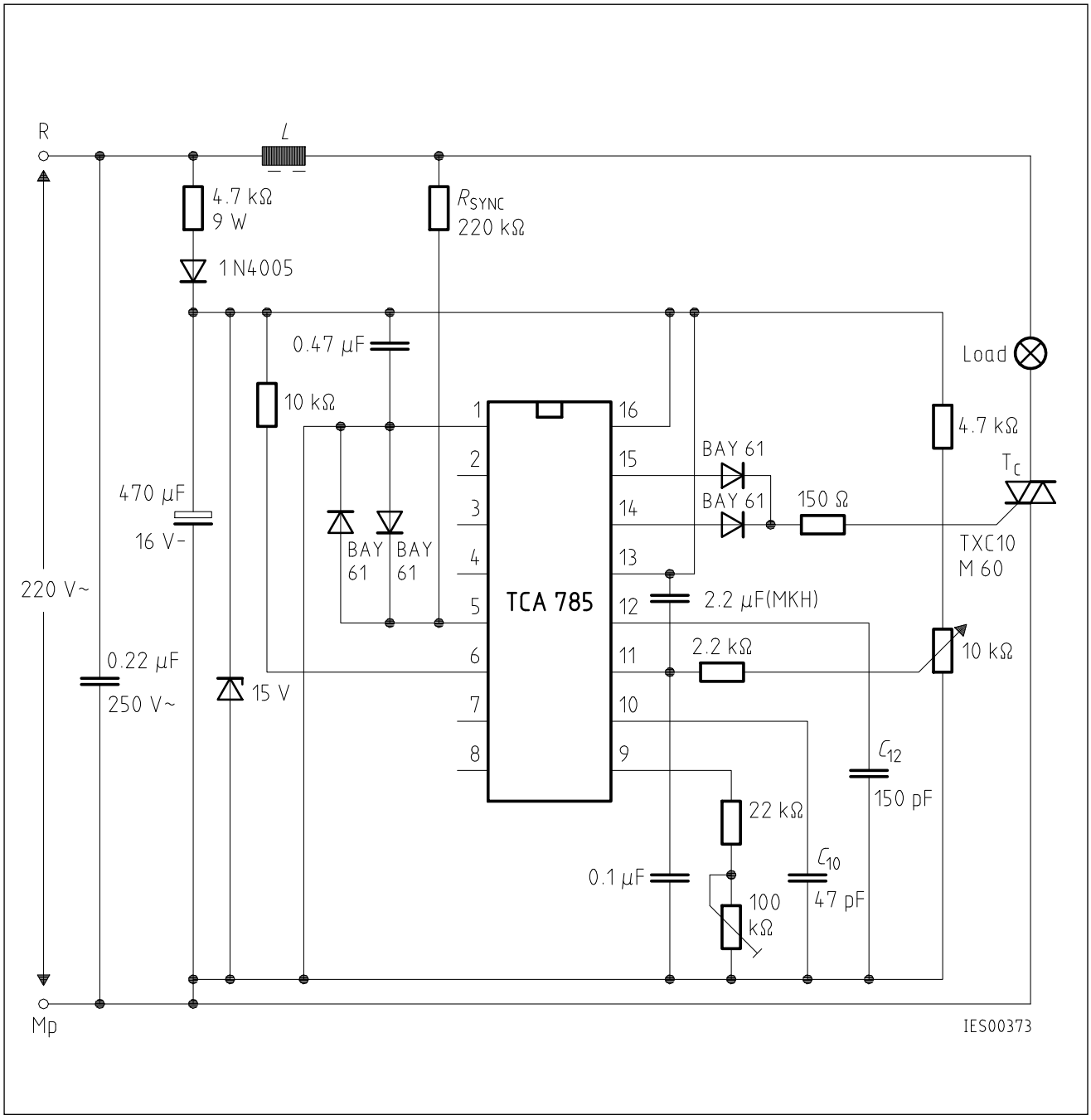
Long Pulse 13



Pulse Extension 12

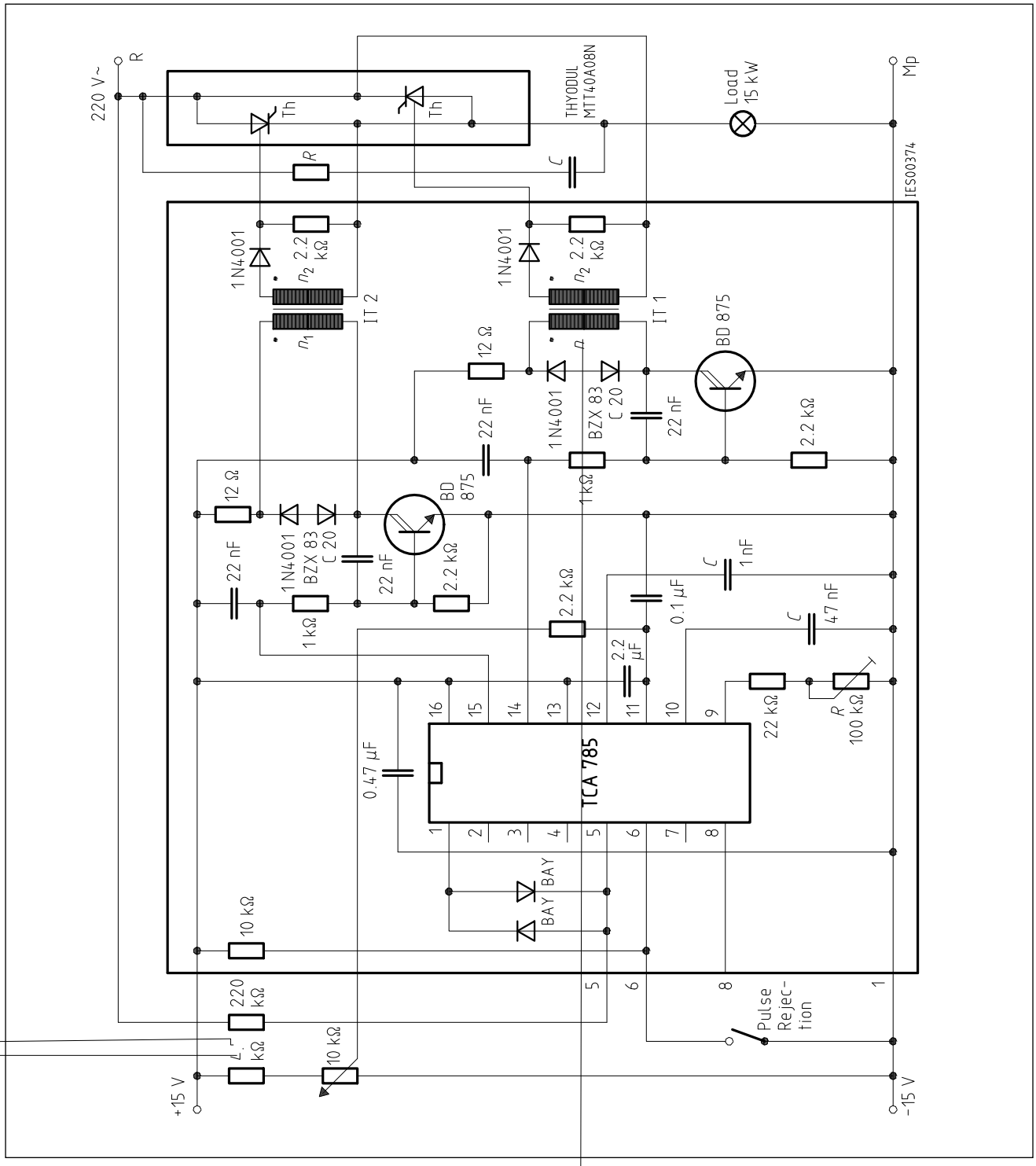


Reference Voltage 8



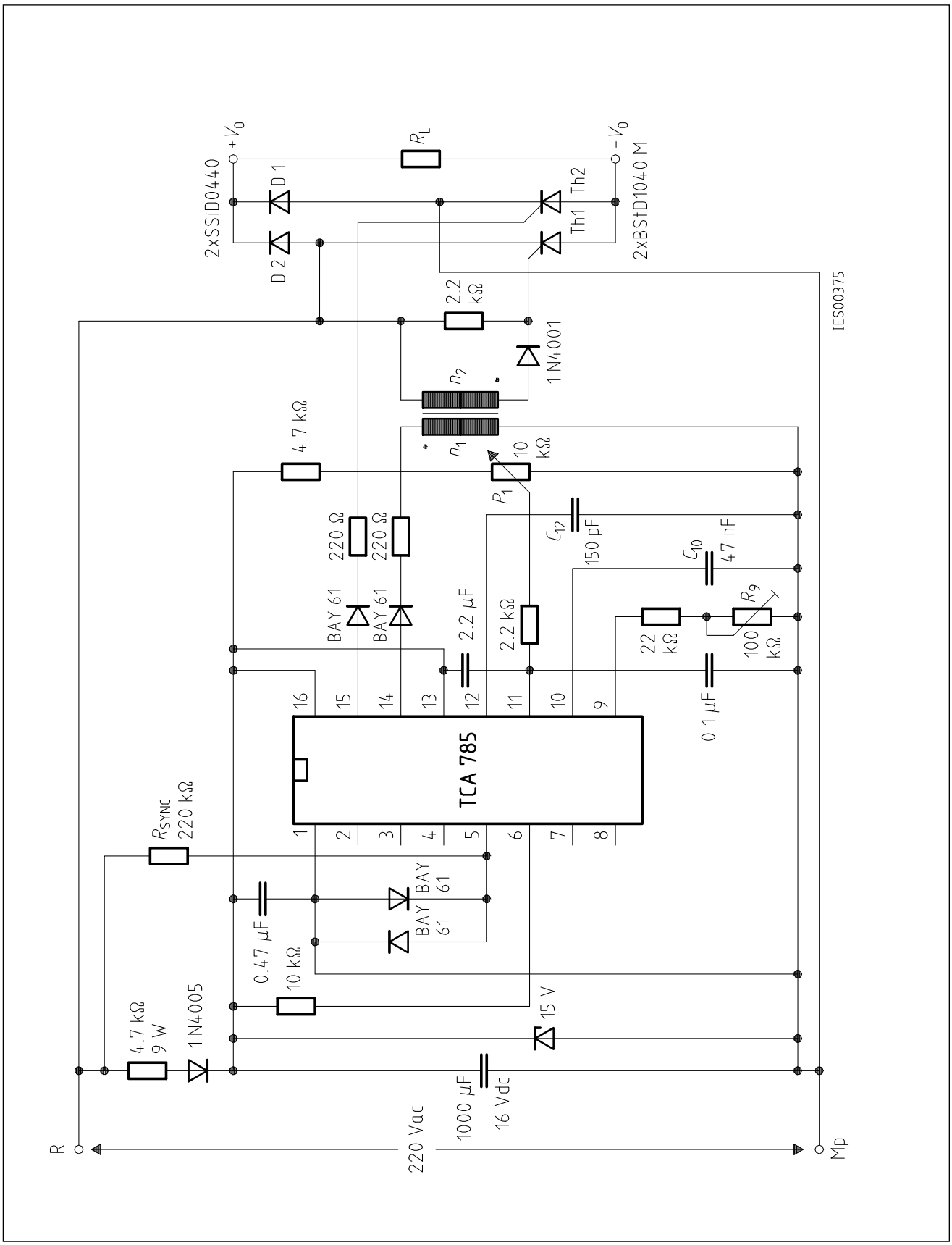
Application Examples
Triac Control for up to 50 mA Gate Trigger Current

A phase control with a directly controlled triac is shown in the figure. The triggering angle of the triac can be adjusted continuously between 0° and 180° with the aid of an external potentiometer. During the positive half-wave of the line voltage, the triac receives a positive gate pulse from the IC output pin 15. During the negative half-wave, it also receives a positive trigger pulse from pin 14. The trigger pulse width is approx. 100 µs.

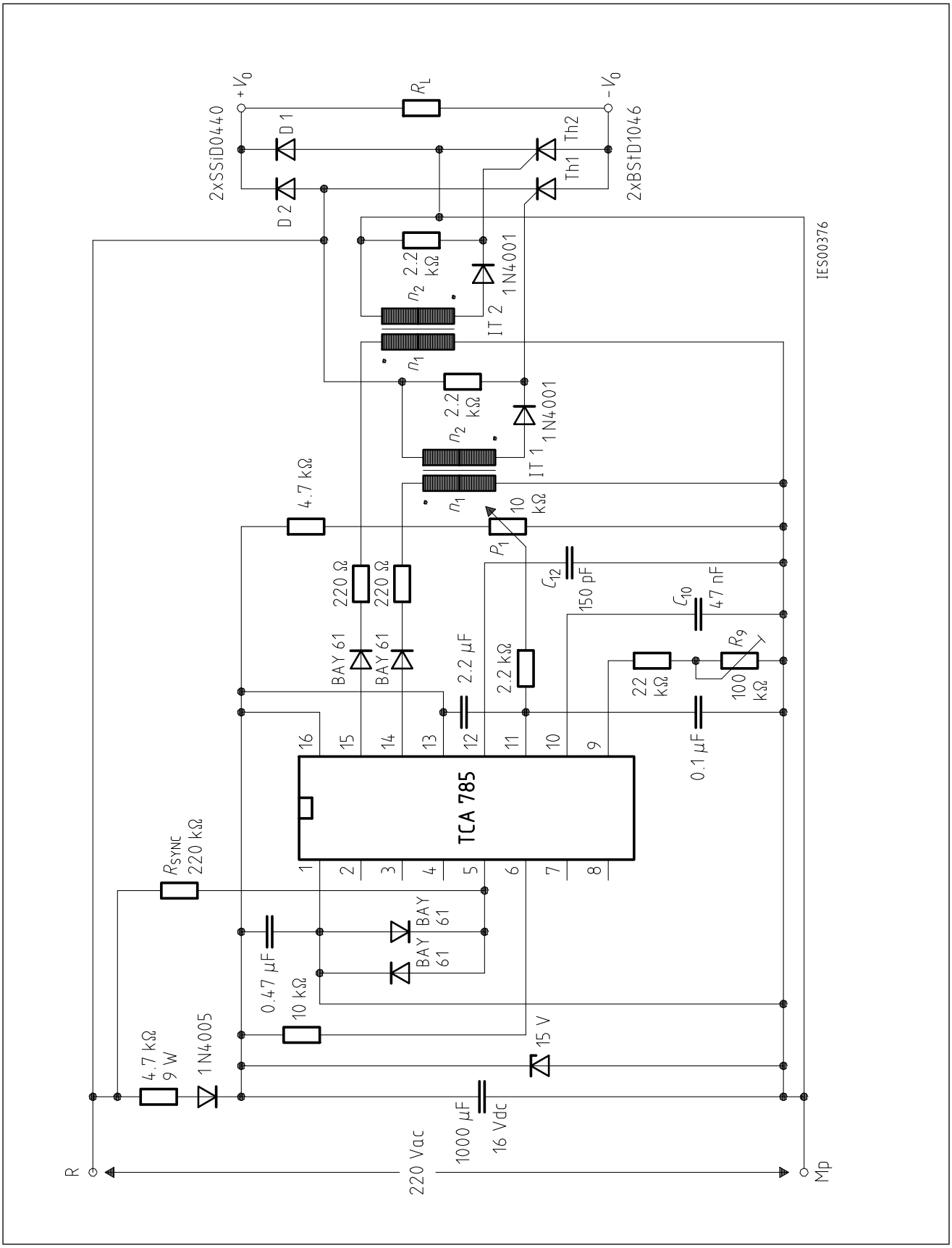


Fully Controlled AC Power Controller Circuit for Two High-Power Thyristors

Shown is the possibility to trigger two antiparalleled thyristors with one IC TCA 785. The trigger pulse can be shifted continuously within a phase angle between 0˚ and 180˚ by means of a potentiometer. During the negative line half-wave the trigger pulse of pin 14 is fed to the relevant thyristor via a trigger pulse transformer. During the positive line half-wave, the gate of the second thyristor is triggered by a trigger pulse transformer at pin 15.



Half-Controlled Single-Phase Bridge Circuit with Trigger Pulse Transformer and Direct Control for Low-Power Thyristors



Half-Controlled Single-Phase Bridge Circuit with Two Trigger Pulse Transformers for Low-Power Thyristors

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